

Earth Engine Dataset Cheat Sheet

Every ID on this page was checked against the live Earth Engine catalog in August 2026 and used to produce real numbers. Copy IDs from *here* — never from memory, and never from whatever an AI assistant suggests.

WHY THIS PAGE EXISTS

AI assistants invent dataset IDs that look completely plausible. They also copy old IDs from tutorials that have since been superseded. A wrong ID is the single most common reason your code will fail in this course. **The fix is mechanical: paste the ID from this sheet into your prompt.**

1 Imagery — the raw material

PURPOSE	DATASET ID	BANDS YOU WILL ACTUALLY USE
Sentinel-2 10 m, every 5 days	COPERNICUS/S2_SR_HARMONIZED	B2 blue · B3 green · B4 red · B8 NIR · B11 / B12 SWIR · SCL scene classification (for cloud masking) Values are scaled *10000 — divide by 10000 for reflectance.
Landsat 9 30 m, every 16 days	LANDSAT/LC09/C02/T1_L2	SR_B2..SR_B7 , QA_PIXEL Use when you need history before 2015 (Sentinel-2 does not exist before then).

2 Ready-made maps — someone already did the hard part

PURPOSE	DATASET ID	NOTES
Land cover 10 m, year 2021	ESA/WorldCover/v200 image: ESA/WorldCover/v200/2021	Band Map . Classes: 10 Tree cover · 20 Shrubland · 30 Grassland · 40 Cropland · 50 Built-up · 60 Bare · 70 Snow/ice · 80 Water · 90 Herbaceous wetland · 95 Mangroves · 100 Moss/lichen
Land cover, near-real-time 10 m, 2015–now	GOOGLE/DYNAMICWORLD/V1	An ImageCollection — filter by date, then <code>.mode()</code> on band <code>label</code> . Classes 0–8: water, trees, grass, flooded_vegetation, crops, shrub_and_scrub, built, bare, snow_and_ice
Forest loss 30 m, 2000–2025	UMD/hansen/global_forest_change_2025_v1_13	<code>treecover2000 (%)</code> · <code>loss (0/1)</code> · <code>lossyear (1–25 → 2001–2025)</code> · <code>gain</code> The version number changes every year. Tutorials citing <code>2024_v1_12</code> or earlier are out of date.

3 Carbon and biomass – read the units twice

PURPOSE	DATASET ID	NOTES
Carbon density 300 m, year 2010	NASA/ORNL/biomass_carbon_density/v1	This is an ImageCollection with one image, not an Image. Use <code>ee.ImageCollection(...).first()</code> . Bands <code>agb</code> , <code>bgb</code> – already in Mg C/ha (carbon) . Do not multiply by 0.47.
Biomass from lidar 1 km, 2019–2021	LARSE/GEDI/GEDI04_B_002	Band <code>MU</code> = mean aboveground biomass density in Mg/ha (biomass, not carbon) . Multiply by 0.47 to get carbon. Covers only where GEDI flew – about 60% of a typical Thai province.

THE UNIT TRAP – THIS WILL BE ON THE TEST

These two datasets look interchangeable and are not. ORNL `agb` is **already carbon**; GEDI `MU` is **biomass**. AI assistants routinely apply the 0.47 carbon fraction to both, which silently halves your ORNL answer. Ask yourself every time: **is this number biomass or carbon?**

4 Fire

PURPOSE	DATASET ID	NOTES
Active fire 1 km, daily	FIRMS	Bands <code>T21</code> (brightness temperature, K), <code>confidence</code> (%). Tells you <i>where something was burning at the moment the satellite passed over</i> – not total area burned.
Burned area 500 m, monthly	MODIS/061/MCD64A1	Band <code>BurnDate</code> : 0 = unburned, 1–366 = day of year it burned. This is the dataset to use when someone asks “ <i>how many hectares burned?</i> ”
Burn severity computed by you	from Sentinel-2	<code>NBR</code> = <code>normalizedDifference(['B8', 'B12'])</code> , then <code>dNBR</code> = <code>NBR_{before} - NBR_{after}</code> . Always mask <code>dNBR</code> to a real burn scar (MCD64A1), or deciduous leaf-drop will register as fire.

5 Climate

PURPOSE	DATASET ID	NOTES
Rainfall 5.5 km, daily, 1981–now	UCSB-CHG/CHIRPS/DAILY	Band <code>precipitation</code> , mm/day. Use <code>.sum()</code> over a date range for a monthly total.
Temperature 11 km, monthly, 1950–now	ECMWF/ERA5_LAND/MONTHLY_AGGR	<code>temperature_2m</code> in Kelvin – subtract 273.15 for °C. <code>total_precipitation_sum</code> in metres – multiply by 1000 for mm.

6 Boundaries — how to define your study area in one line

```
# Any Thai province, no shapefile upload needed
aoi = (ee.FeatureCollection('FAO/GAUL/2015/level1')
      .filter(ee.Filter.eq('ADM0_NAME', 'Thailand'))
      .filter(ee.Filter.eq('ADM1_NAME', 'Nan'))) # <- change this

# level2 = district (amphoe). Field name is ADM2_NAME.
```

Province names are spelled in English exactly as: Nan , Chiang Mai , Chiang Rai , Nakhon Ratchasima , Kanchanaburi , Tak , Mae Hong Son , Phrae , Lampang , Uttaradit . There are 76 in the dataset. To list them all:

```
print(aoi_all.aggregate_array('ADM1_NAME').getInfo())
```

7 Numbers to sanity-check against

These were computed with the code in this course, so if your output is wildly different from the same area, something is wrong.

QUANTITY	AREA	VALUE	SOURCE
Province area	Nan	1,227,687 ha	FAO GAUL
Province area	Chiang Mai	2,225,886 ha	FAO GAUL
Mean NDVI, dry season	Nan	0.75	Sentinel-2
Tree cover	Nan	1,033,493 ha (84%)	ESA WorldCover
Tree cover	Nan	1,085,511 ha (88%)	Dynamic World
Mean AGB carbon	Nan forest	46 Mg C/ha	ORNL 2010
Mean AGB biomass	Nan forest	163 Mg/ha	GED1 2019–21
Burned area, Jan–May	Chiang Mai	67,128 ha (2024)	MCD64A1
Burned area, Jan–May	Chiang Mai	33,482 ha (2025)	MCD64A1
Rainfall, Jan–Mar	Chiang Mai	20 mm	CHIRPS 2024
Rainfall, Jul–Sep	Chiang Mai	770 mm	CHIRPS 2024

QUOTA — KEEP YOUR PROJECT ALIVE

Your free Earth Engine account gets **150 EECU-hours per month**. That is plenty for this course *if* you follow two rules: keep your study area to **one province or smaller**, and always pass an explicit `scale=` to `reduceRegion` (use `100` for area statistics, not `10`). Asking for a whole country at 10 m resolution can burn a month of quota in one cell.